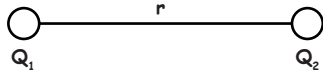


ELECTROSTATIC POTENTIAL ENERGY

2 point charges

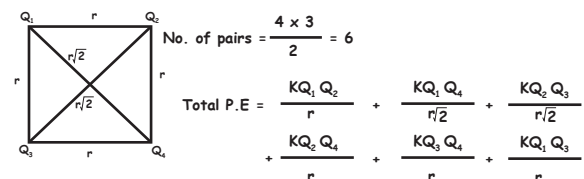


$$\Delta U = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r}$$

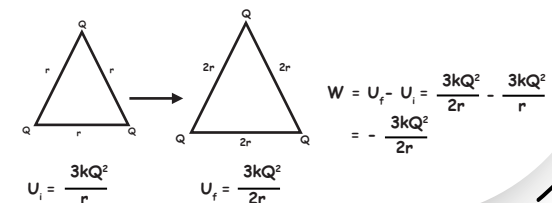
like charges - positive (repulsive energy)
Unlike charges - negative (attractive energy)

System of charges

$$\Delta U_{\text{system}} = \sum \Delta U_{\text{pair}} \quad \text{No. of pairs} = \frac{n(n-1)}{2}$$



WORK DONE IN REARRANGEMENT OF THE SYSTEM



ELECTROSTATIC POTENTIAL

$$V_p = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

$$V_{AB} = V_B - V_A = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$

WORK DONE

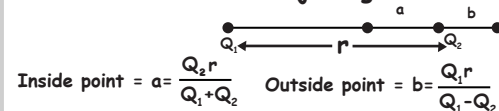
$$W = q[V_B - V_A]$$

$$W = 1 \times (-500 - 500) = -1000J$$

Superposition of potential - Algebraic sum of all potentials

ZERO POTENTIAL

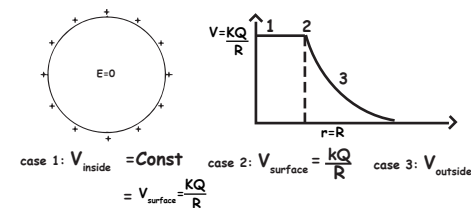
- a) Like charge - no zero potential point
b) Unlike charge - 2 points of zero potential on line joining



$$d_{\min} = \frac{2kqQ}{mv^2}$$

Large distance fixed

POTENTIAL OF CHARGED CONDUCTING SPHERE

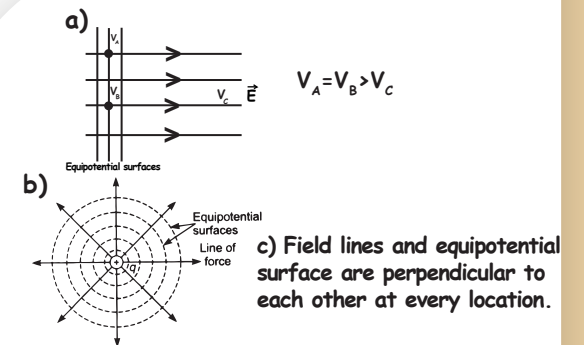


Redistribution of Charge when two Conducting sphere are connected
Charge flow from higher potential to lower potential
Final potentials of spheres are equal

$$\frac{Q_1}{r_1} = \frac{Q_2}{r_2} \Rightarrow \frac{Q_1}{Q_2} = \frac{r_1}{r_2}$$

$$\frac{Q_1}{Q_2} = \frac{r_1}{r_2} \Rightarrow \frac{E_1}{E_2} = \frac{r_2}{r_1} \Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$$

EQUIPOTENTIAL SURFACE



d) work done in moving a charge on equipotential surface is 0

ELECTRIC FIELD & POTENTIAL

$$E = -\frac{dV}{dr} \quad E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

$$\Delta V = -\int \vec{E} \cdot d\vec{r}$$

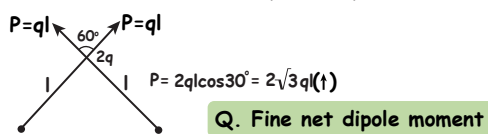
ELECTROSTATICS-02

DIPOLE

Dipole moment

$$\vec{P} = q \vec{2l}$$

Direction is from -Q to +Q



ELECTRIC FIELD

$$E_p = \frac{Kp}{r^3} \sqrt{3\cos^2\theta + 1}$$

$E_{\text{axial}} = \frac{Kp}{r^3} \sqrt{3+1} = \frac{2Kp}{r^3}$ 0° with dipole moment
 E_{net} from -Q to +Q

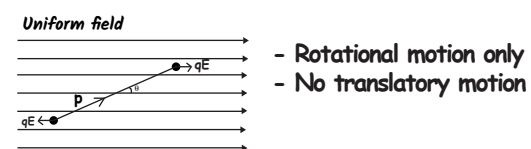
$E_{\text{equatorial}} = \frac{Kp}{r^3} \sqrt{0+1} = \frac{Kp}{r^3}$ 180° with dipole moment
 E_{net} from +Q to -Q

ELECTRIC POTENTIAL

$$V_p = \frac{kp \cos\theta}{r^2}$$

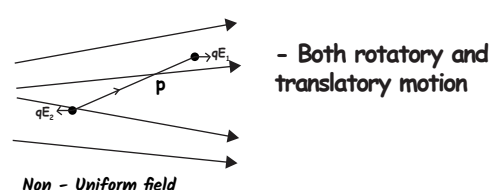
$$V_{\text{axial}} = \frac{Kp}{r^2} \quad V_{\text{equatorial}} = 0$$

TORQUE



$$\vec{\tau} = \vec{P} \times \vec{E}$$

$$\tau = PES \sin\theta$$



$$\tau_{\max} = pE \sin\theta = pE, (\theta = 90^\circ)$$

$$\tau_{\min} = pE \sin\theta = 0, (\theta = 0^\circ)$$

WORK DONE

$$W = pE (\cos\theta_1 - \cos\theta_2)$$

POTENTIAL ENERGY

$$U = -pE \cos\theta$$

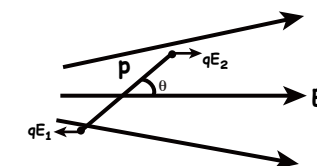
1) $U_{\min}$ (stable equilibrium)

$$\theta = 0^\circ \quad \cos\theta = 1 \quad U = -pE$$

2) $U_{\max}$ (Unstable equilibrium)

$$\theta = 180^\circ \quad \cos\theta = -1 \quad U = +pE$$

Force in Non-Uniform field



$$F_x = p \frac{dE}{dx} \cos\theta$$

where, dE = small change in field at the two locations of the charges